

SEPHORA – DATABASE MANAGEMENT SYSTEM

PRODUCT REVIEW AND RATING MANAGEMENT SYSTEM

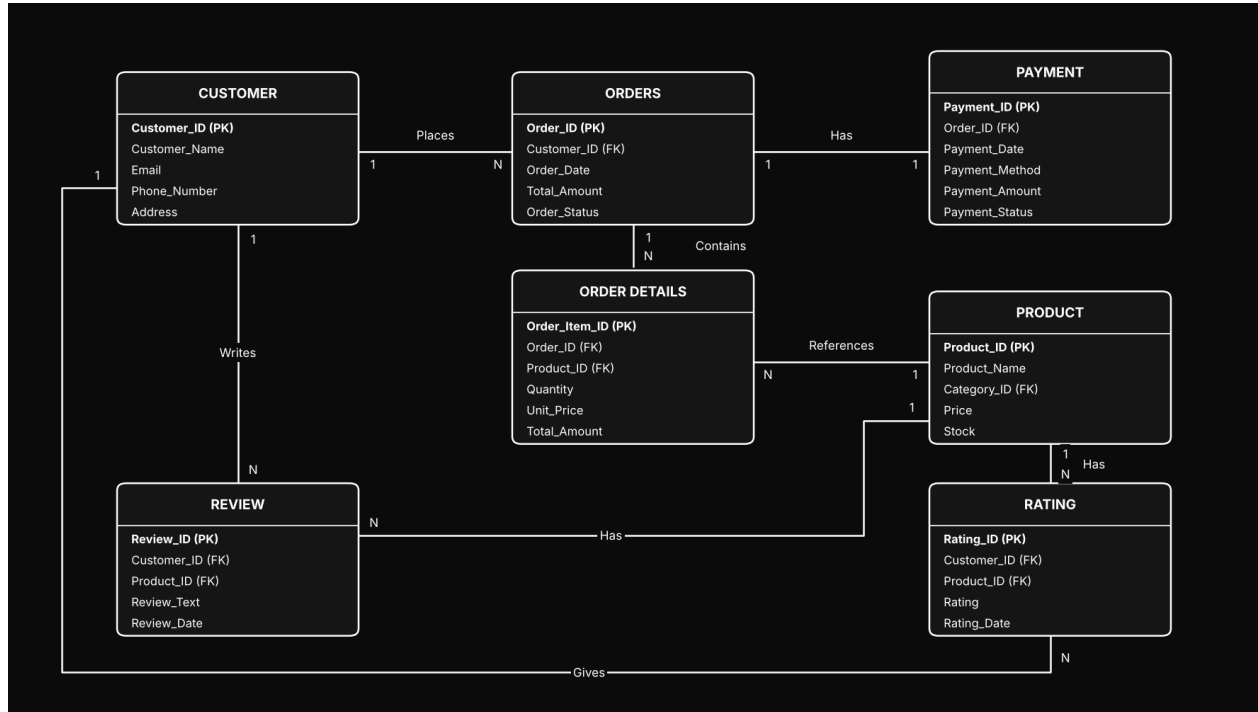
1. REVIEW TABLE

Attribute	Data Type	Description
Review_ID	NUMBER	Unique review ID – Primary Key
Customer_ID	NUMBER	Customer who submitted the review – Foreign Key
Product_ID	NUMBER	Product being reviewed – Foreign Key
Review_Text	VARCHAR2(200)	Customer's review/comment
Review_Date	DATE	Date of the review

2. RATING TABLE

Column	Data Type	Description
Rating_ID	NUMBER	Unique ID for each rating – Primary Key
Customer_ID	NUMBER	Customer who gives the rating – Foreign Key
Product_ID	NUMBER	Product being rated – Foreign Key
Rating	NUMBER(1)	Rating given by customer from 1 to 5
Rating_Date	DATE	Date of the rating

3. RELATIONSHIP DIAGRAM



4. TABLE CREATION

4.1 CREATE REVIEW TABLE

```
SQL> CREATE TABLE SephoraReview (  
2     Review_ID NUMBER PRIMARY KEY,  
3     Customer_ID NUMBER REFERENCES SephoraCustomer(Customer_ID),  
4     Product_ID NUMBER REFERENCES SephoraProduct(Product_ID),  
5     Review_Text VARCHAR2(200),  
6     Review_Date DATE  
7 );
```

Table created.

```
SQL> DESC SephoraReview;
```

Name	Null?	Type
REVIEW_ID	NOT NULL	NUMBER
CUSTOMER_ID		NUMBER
PRODUCT_ID		NUMBER
REVIEW_TEXT		VARCHAR2(200)
REVIEW_DATE		DATE

4.2 CREATE RATING TABLE

```
SQL> CREATE TABLE SephoraRating (  
2     Rating_ID NUMBER PRIMARY KEY,  
3     Customer_ID NUMBER REFERENCES SephoraCustomer(Customer_ID),  
4     Product_ID NUMBER REFERENCES SephoraProduct(Product_ID),  
5     Rating NUMBER(1) CHECK (Rating BETWEEN 1 AND 5),  
6     Rating_Date DATE  
7 );
```

Table created.

```
SQL> DESC SephoraRating;
```

Name	Null?	Type
RATING_ID	NOT NULL	NUMBER
CUSTOMER_ID		NUMBER
PRODUCT_ID		NUMBER
RATING		NUMBER(1)
RATING_DATE		DATE

5. REVIEW DATA INSERTION

```
SQL> INSERT INTO SephoraReview VALUES
  2  (1, 1, 201, 'Long lasting matte lipstick', TO_DATE('01-10-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO SephoraReview VALUES
  2  (2, 2, 202, 'Serum gives a nice glow', TO_DATE('02-10-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO SephoraReview VALUES
  2  (3, 3, 203, 'Very good moisturizer', TO_DATE('02-10-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO SephoraReview VALUES
  2  (4, 1, 204, 'Light sunscreen, no white cast', TO_DATE('03-10-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO SephoraReview VALUES
  2  (5, 2, 205, 'Mascara does not smudge', TO_DATE('03-10-2026','DD-MM-YYYY'));

1 row created.

SQL> INSERT INTO SephoraReview VALUES
  2  (6, 3, 206, 'Gentle face wash for my skin', TO_DATE('04-10-2026','DD-MM-YYYY'));

1 row created.

SQL> COMMIT;

Commit complete.
```

5.1 REVIEW TABLE DISPLAY

```
SQL> SELECT * FROM SephoraReview;
```

REVIEW_ID	CUSTOMER_ID	PRODUCT_ID	REVIEW_TEXT	REVIEW_DA
1	1	201	Long lasting matte lipstick	01-OCT-26
2	2	202	Serum gives a nice glow	02-OCT-26
3	3	203	Very good moisturizer	02-OCT-26
4	1	204	Light sunscreen, no white cast	03-OCT-26
5	2	205	Mascara does not smudge	03-OCT-26
6	3	206	Gentle face wash for my skin	04-OCT-26

```
6 rows selected.
```

6. RATING DATA INSERTION

```
SQL> INSERT INTO SephoraRating VALUES  
2 (1, 1, 201, 5, TO_DATE('01-10-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL> INSERT INTO SephoraRating VALUES  
2 (2, 2, 202, 4, TO_DATE('02-10-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL> INSERT INTO SephoraRating VALUES  
2 (3, 3, 203, 5, TO_DATE('02-10-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL> INSERT INTO SephoraRating VALUES  
2 (4, 1, 204, 4, TO_DATE('03-10-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL> INSERT INTO SephoraRating VALUES  
2 (5, 2, 205, 5, TO_DATE('03-10-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL> INSERT INTO SephoraRating VALUES  
2 (6, 3, 206, 4, TO_DATE('04-10-2026','DD-MM-YYYY'));
```

```
1 row created.
```

```
SQL> COMMIT;
```

```
Commit complete.
```

6.1 RATING TABLE DISPLAY

```
SQL> SELECT * FROM SephoraRating;
```

RATING_ID	CUSTOMER_ID	PRODUCT_ID	RATING	RATING_DA
1	1	201	5	01-OCT-26
2	2	202	4	02-OCT-26
3	3	203	5	02-OCT-26
4	1	204	4	03-OCT-26
5	2	205	5	03-OCT-26
6	3	206	4	04-OCT-26

6 rows selected.

7. RETRIEVE PRODUCT REVIEW DETAILS

```
SQL> SELECT
2     r.Review_ID,
3     c.Customer_Name,
4     p.Product_ID,
5     p.Product_Name,
6     r.Review_Text,
7     r.Review_Date
8 FROM SephoraReview r
9 JOIN SephoraCustomer c
10      ON r.Customer_ID = c.Customer_ID
11 JOIN SephoraProduct p
12      ON r.Product_ID = p.Product_ID
13 ORDER BY r.Review_Date DESC;
```

REVIEW_ID	CUSTOMER_NAME	PRODUCT_ID	PRODUCT_NAME	REVIEW_TEXT	REVIEW_DA
6	Kavya	206	Gentle Face Wash	Gentle face wash for my skin	04-OCT-26
5	Meena	205	Waterproof Mascara	Mascara does not smudge	03-OCT-26
4	Priya	204	SPF 50 Sunscreen	Light sunscreen, no white cast	03-OCT-26
3	Kavya	203	Hydrating Moisturizer	Very good moisturizer	02-OCT-26
2	Meena	202	Vitamin C Serum	Serum gives a nice glow	02-OCT-26
1	Priya	201	Matte Liquid Lipstick	Long lasting matte lipstick	01-OCT-26

6 rows selected.

8. CALCULATE AVERAGE PRODUCT RATINGS

```
SQL> SELECT
  2     Product_ID,
  3     AVG(Rating) AS Average_Rating
  4 FROM SephoraRating
  5 GROUP BY Product_ID
  6 ORDER BY Product_ID;
```

PRODUCT_ID	AVERAGE_RATING
201	5
202	4
203	5
204	4
205	5
206	4

6 rows selected.

9. FILTER PRODUCTS BY AVERAGE RATING

```
SQL> SELECT
  2     Product_ID,
  3     AVG(Rating) AS Average_Rating
  4 FROM SephoraRating
  5 GROUP BY Product_ID
  6 HAVING AVG(Rating) >= 4
  7 ORDER BY Average_Rating DESC;
```

PRODUCT_ID	AVERAGE_RATING
201	5
203	5
205	5
202	4
204	4
206	4

6 rows selected.

10. FILTER PRODUCTS BY AVERAGE RATING WITH PRODUCT NAME

```
SQL> COLUMN Average_Rating FORMAT 9.99
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     AVG(r.Rating) AS Average_Rating
  5 FROM SephoraRating r
  6 JOIN SephoraProduct p
  7     ON r.Product_ID = p.Product_ID
  8 GROUP BY
  9     p.Product_ID,
10     p.Product_Name
11 HAVING AVG(r.Rating) >= 4
12 ORDER BY Average_Rating DESC;
```

PRODUCT_ID	PRODUCT_NAME	AVERAGE_RATING
201	Matte Liquid Lipstick	5.00
203	Hydrating Moisturizer	5.00
205	Waterproof Mascara	5.00
202	Vitamin C Serum	4.00
204	SPF 50 Sunscreen	4.00
206	Gentle Face Wash	4.00

6 rows selected.

11. PRODUCT RATING REPORT

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     COUNT(r.Rating_ID) AS Total_Ratings,
  5     ROUND(AVG(r.Rating), 2) AS Average_Rating,
  6     MAX(r.Rating) AS Highest_Rating,
  7     MIN(r.Rating) AS Lowest_Rating
  8 FROM SephoraRating r
  9 JOIN SephoraProduct p
10     ON r.Product_ID = p.Product_ID
11 GROUP BY p.Product_ID, p.Product_Name
12 ORDER BY Average_Rating DESC;
```

PRODUCT_ID	PRODUCT_NAME	TOTAL_RATINGS	AVERAGE_RATING	HIGHEST_RATING	LOWEST_RATING
201	Matte Liquid Lipstick	1	5	5	5
203	Hydrating Moisturizer	1	5	5	5
205	Waterproof Mascara	1	5	5	5
202	Vitamin C Serum	1	4	4	4
204	SPF 50 Sunscreen	1	4	4	4
206	Gentle Face Wash	1	4	4	4

6 rows selected.